

MLB HITTERS

SALARY PREDICTION REGRESSION MODEL



GROUP 6

ST 3008 Applied Statistical Models

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Introduction & Literature Review

1.Introduction

The “Hitters” dataset is a widely used resource for regression analysis and contains the records of 322 baseball players from the Major league baseball (MLB) 1986-1987 season. This dataset was originally taken from the StatLib library which is maintained by Carnegie Mellon University. The salary data was obtained from the Sports Illustrated magazine April 20, 1987 issue, while the career statistics were obtained from the 1987 Baseball Encyclopedia Update published by Collier Books, Macmillan Publishing Company, New York. This dataset is also a part of the R-package ISLR and is used in the related book by G. James et al. (2013) "An Introduction to Statistical Learning with applications in R" to demonstrate how Ridge regression and LASSO are performed using R.

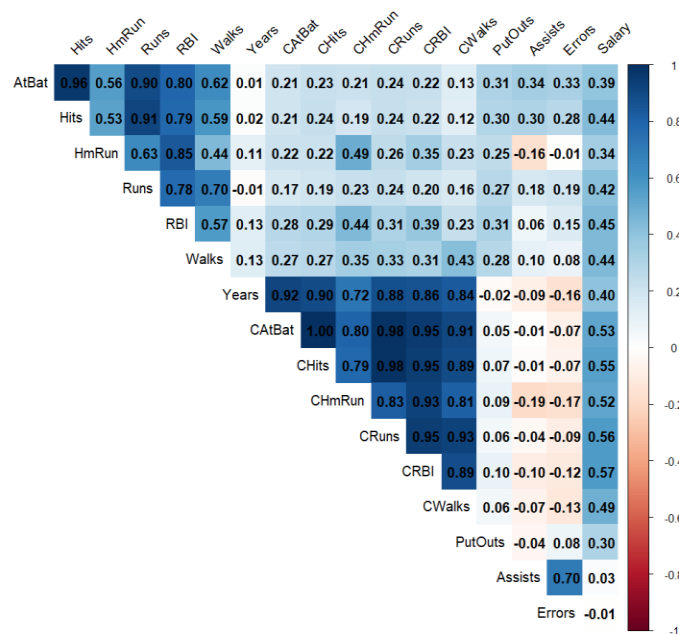
This dataset includes 20 variables including player performance metrics (from the 1986 season) that are commonly used in baseball such as hits, runs as well as player career metrics such as career hits and career runs and the leagues and divisions in which the players played during the 1986 season. The dataset also contains the player salary information for the 1987 season and thus provides a good foundation to enable prediction of player salaries based on key player performance metrics and career statistics.

This review looks at the use of this dataset in various literature and the and aims to understand the analyses that have been done on the dataset as well as the major findings of those studies and provides a brief review of these studies.

2.Use in other literature

This dataset is primarily used to build models for the prediction of the salary of players using various player performance metrics. The analyses on this dataset are primarily found on the dataset sharing platform “Kaggle” and it is also available in the R package ISLR and is used as an example dataset to demonstrate Ridge and Lasso regression in R in the book "An Introduction to Statistical Learning with applications in R" by G. James et al. (2013). This is a part of the data that was used in the 1988 ASA Graphics Section Poster Session.

A primary initial step in a majority of the analysis done in the Kaggle notebooks is explanatory data analysis, which uses plots and visualizations to identify relationships between the variables and detect any issues in model building and fitting. A key feature noticed in the explanatory data analysis done in these studies is that there is strong correlations between quite a few of the variables. This is shown in the image below which shows a correlation matrix of the variables where the dark shades of blue represent strong correlations and dark shades of red represent strong negative correlations.



"An Introduction to Statistical Learning with applications in R" discusses the use of subset selection to select predictors for the model using techniques such as forward, backward or stepwise selection. Techniques such as these help to reduce overfitting.

Another set of useful techniques that have been discussed in the book and used in a wide range of the Kaggle notebooks are Ridge regression and Lasso regression.

- I. Ridge regression: Ridge regression is a linear regression technique used to address multicollinearity and overfitting in models with highly correlated independent variables. It adds a penalty term to the ordinary least squares (OLS) cost function, effectively shrinking the coefficients towards zero and reducing their variance.
- II. Lasso regression: Lasso regression, also known as Least Absolute Shrinkage and Selection Operator, is a type of linear regression that uses a regularization technique to shrink the coefficients of less important variables towards zero, potentially making them exactly zero.

The book and the Kaggle notebooks also discuss more advanced tree-based methods such as Random Forest models and other more advanced machine learning techniques such as boosting in the form of XgBoost models, and other machine learning models such as Support vector regression and K-Nearest Neighbors.

- I. Random forest regression: Random Forest Regression is an ensemble machine learning method that leverages multiple decision trees to predict a continuous dependent variable. It operates by constructing a "forest" of numerous decision trees during training and then

averaging the individual predictions of these trees to produce a more accurate and stable final prediction.

- II. XgBoost: XGBoost is a powerful and efficient gradient boosting library that implements machine learning algorithms under the gradient boosting framework.
- III. Support vector regression: Support Vector Regression (SVR) is a powerful supervised learning algorithm that is a variation of the more widely known Support Vector Machine (SVM). While SVMs are primarily used for classification, SVR is adapted for regression tasks, meaning it's used to predict continuous numerical values.
- IV. K-Nearest Neighbors: A non-parametric method that predicts a player's salary based on the average salary of the 'k' most similar players in the dataset.

3. Major Findings and review

The presence of strongly correlated variables in this dataset is highlighted in a majority of the studies conducted using it. This is compensated for by applying more advanced regression and machine learning techniques. Furthermore, offensive statistics of the players such as hits and career runs are found to be strong predictors of salary. **Subset selection** and **Lasso regression** are thus found to be important tools in order to select these significant predictors efficiently.

The explanatory data analysis conducted in most of the studies also shows that the salary response is heavily right skewed implying the necessity of applying a log transformation to the salary variable when fitting a model. As the more advanced regression techniques lie outside the scope of this study it is necessary to focus on methods used in classical regression to ensure an adequate model is fitted despite the challenges posed by the collinearity of the variables.

Descriptive Analysis

1. Overview of Variables

The dataset contains information on 322 Major League Baseball players, described by 20 variables. Out of these, 16 are numeric, 3 are categorical variables, and 1 is the response variables (Salary).

A detailed summary of the dataset variables is given below:

Variable	Type	Description
AtBat	Numeric	Number of times at bat in 1986
Hits	Numeric	Number of hits in 1986
HmRun	Numeric	Number of home runs in 1986
Runs	Numeric	Number of runs scored in 1986
RBI	Numeric	Runs batted in during 1986
Walks	Numeric	Number of walks in 1986
Years	Numeric	Number of years in the Major Leagues
CAtBat	Numeric	Career times at bat up to 1986
CHits	Numeric	Career hits up to 1986

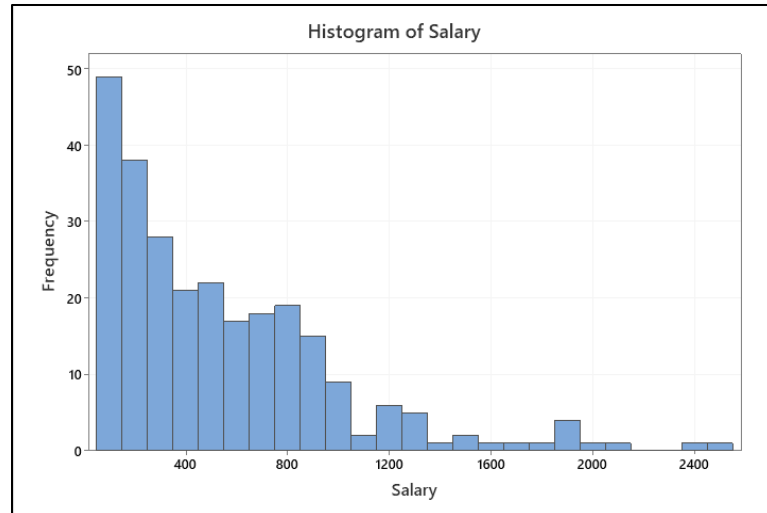
CHmRun	Numeric	Career home runs up to 1986
CRuns	Numeric	Career runs up to 1986
CRBI	Numeric	Career runs batted in up to 1986
CWalks	Numeric	Career walks up to 1986
League	Categorical	League in 1986(A=American, N = National)
Division	Categorical	Division in 1986 (E=East, W=West)
PutOuts	Numeric	Put outs in 1986
Assists	Numeric	Assists in 1986
Errors	Numeric	Errors in 1986
Salary	Numeric (Response)	1987 annual salary (in \$1000s), with 59 missing values
NewLeague	Categorical	League at the beginning of 1987 (A = American, N = National)

2.Summary Statistics

Statistics

Variable	N	N*	Mean	StDev	Minimum	Q1	Median	Q3	Maximum
AtBat	322	0	380.93	153.40	16.00	254.75	379.50	512.25	687.00
Hits	322	0	101.02	46.45	1.00	63.75	96.00	137.25	238.00
HmRun	322	0	10.770	8.709	0.000	4.000	8.000	16.000	40.000
Runs	322	0	50.91	26.02	0.00	30.00	48.00	69.25	130.00
RBI	322	0	48.03	26.17	0.00	28.00	44.00	65.00	121.00
Walks	322	0	38.74	21.64	0.00	22.00	35.00	53.25	105.00
Years	322	0	7.444	4.926	1.000	4.000	6.000	11.000	24.000
CAtBat	322	0	2649	2324	19	813	1928	3932	14053
CHits	322	0	717.6	654.5	4.0	208.0	508.0	1063.0	4256.0
CHmRun	322	0	69.49	86.27	0.00	13.75	37.50	90.50	548.00
CRuns	322	0	358.8	334.1	1.0	99.8	247.0	532.8	2165.0
CRBI	322	0	330.1	333.2	0.0	87.5	220.5	428.5	1659.0
CWalks	322	0	260.2	267.1	0.0	66.8	170.5	342.8	1566.0
PutOuts	322	0	288.9	280.7	0.0	109.0	212.0	325.0	1378.0
Assists	322	0	106.91	136.85	0.00	7.00	39.50	167.00	492.00
Errors	322	0	8.040	6.368	0.000	3.000	6.000	11.000	32.000
Salary	263	59	535.9	451.1	67.5	190.0	425.0	750.0	2460.0

- Players had an average of 101 hits in 1986, with a maximum of 238 hits.
- The average career hits (CHits) is around 718, with some players exceeding 4000 career hits.



- The salary variable has a wide range, from 67.5 (thousand USD) to 2460 (thousand USD), with a median of 425, indicating a right-skewed distribution.

League

League	Count	Percent
A	175	54.35
N	147	45.65
N=	322	

Division

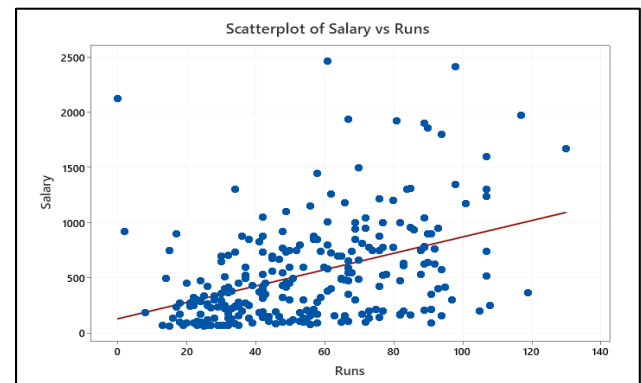
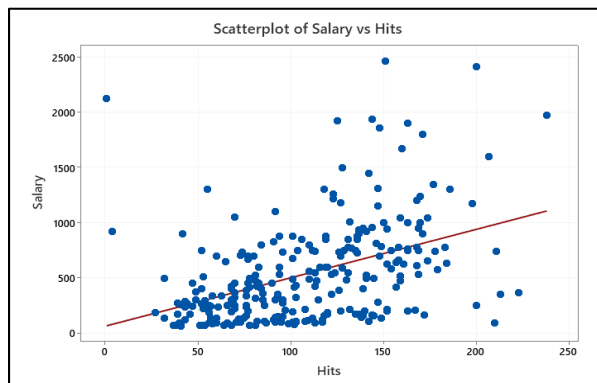
Division	Count	Percent
E	157	48.76
W	165	51.24
N=	322	

NewLeague

NewLeague	Count	Percent
A	176	54.66
N	146	45.34
N=	322	

- The categorical variables are fairly evenly distributed, with slight variations between categories.
- These frequencies help understand how players are spread across leagues and divisions.

3.Bivariate Analysis



- The scatterplot of Salary vs Hits shows a generally increasing trend, indicating that players with more hits tend to have higher salaries. Similarly, the Salary vs Runs plot suggests that higher run scorers are compensated more.

Correlations – Last Season Performance

	Hits	HmRun	Runs	RBI
HmRun	0.562			
Runs	0.922	0.651		
RBI	0.811	0.855	0.798	
Salary	0.439	0.343	0.420	0.449

- Salary shows moderate positive correlations with Hits, Runs, and RBI. Correlation with Home Runs is relatively weaker. Strong correlations exist between the following predictors:
Hits and Runs (0.922)
Hits and RBI (0.811)
Runs and RBI (0.798)
- This indicates that while season performance measures explain Salary to some extent, they are also highly related to each other (possible multicollinearity).

Correlations – Career Performance

	CHits	CHmRun	CRuns	CRBI
CHmRun	0.783			
CRuns	0.985	0.820		
CRBI	0.945	0.929	0.944	
Salary	0.549	0.525	0.563	0.567

- Career statistics show slightly stronger correlations with Salary compared to single season performance. This suggests that long-term consistent performance is more strongly associated with higher salaries than just one season's performance.



- The boxplot of Salary across Divisions shows noticeable differences in distribution. Players in “East” division tend to have higher median salaries compared to the “West” division. The spread of salaries is also wider in East division, indicating greater variability in earnings. This suggests that Division, as a categorical factor, may influence Salary beyond individual performance statistics.

Advanced Analysis

This report presents a comprehensive analysis of baseball player salaries using multiple linear regression modeling. The analysis identifies significant predictors, addresses multicollinearity issues, incorporates categorical variables and interaction effects, and evaluates model adequacy through rigorous diagnostic testing.

A forward selection procedure was implemented to identify the most significant numerical predictors of Salary. The process began by evaluating each predictor individually. Career Runs Batted In (**CRBI**) emerged as the strongest single predictor ($R^2 = 0.3215$, $p < 2.2e-16$), explaining approximately 32.15% of the variance in Salary. Subsequently, predictors were added sequentially based on their ability to significantly improve the model fit, as determined by p-values of ANOVA F-tests. The selection order was as follows: Hits, PutOuts, AtBat, and Walks. This iterative process resulted in an initial main effects model: **Salary ~ CRBI + Hits + PutOuts + AtBat + Walks**.

Variance Inflation Factors (VIFs) were calculated for the initial model to assess the correlation among predictors. The analysis revealed unacceptably high VIF values for both AtBat (15.29) and Hits (14.23), indicating that these variables provided redundant information and were distorting the model's coefficient estimates. To resolve this, AtBat was removed from the model, a decision supported by its higher VIF with Hits. The revised model, **Salary ~ CRBI + Hits + PutOuts + Walks**. This shows VIFs all around 1, confirming that multicollinearity was no longer an issue.

The significance of the categorical variables, League and Division, was evaluated using an overall model F-test method. This statistical test compares a full model (including the categorical variable) to a reduced model (without it) to determine if the additional variable adds significant explanatory power. The variable League was found to be statistically insignificant (F-statistic = 1.22) and was therefore excluded. Then, the Division variable was highly significant (F-statistic = 14.11). Which indicated that players in different Divisions had significantly different salaries. The model was updated to include this factor, **Salary ~ CRBI + Hits + PutOuts + Walks + Division**.

To enhance the model's predictive accuracy, all possible two-way interactions between the selected numerical variables were systematically tested. This was achieved by adding interaction terms one by one and assessing their significance via ANOVA. Three interaction terms were found to be statistically significant, CRBI*Hits, Hits*Walks and CRBI*PutOuts. The addition of these interactions signify that the effect of one predictor on Salary is dependent on the level of another. The final model, which includes these synergistic effects, is represented as, **Salary ~ CRBI + Hits + PutOuts + Walks + Division + CRBIHits + HitsWalks + CRBI*PutOuts**.

Analysis of Variance Table

Response: Salary

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
CRBI	1	17139434	17139434	200.8357	< 2.2e-16 ***
Hits	1	5533119	5533119	64.8358	3.158e-14 ***
PutOuts	1	1397263	1397263	16.3728	6.909e-05 ***
Walks	1	539490	539490	6.3216	0.0125464 *
as.factor(Division)	1	1230539	1230539	14.4192	0.0001831 ***
CRBI:Hits	1	3107825	3107825	36.4167	5.610e-09 ***
Hits:Walks	1	2350979	2350979	27.5482	3.236e-07 ***
CRBI:PutOuts	1	343963	343963	4.0305	0.0457446 *
Residuals	254	21676501	85341		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Call:
lm(formula = Salary ~ CRBI + Hits + PutOuts + Walks + as.factor(Division) +
    CRBI * Hits + Hits * Walks + CRBI * PutOuts)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1010.84	-167.16	-41.31	145.89	1537.34

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.914e+02	9.437e+01	6.267	1.57e-09 ***
CRBI	-6.086e-03	1.555e-01	-0.039	0.968813
Hits	-3.593e+00	9.233e-01	-3.891	0.000128 ***
PutOuts	2.497e-02	1.086e-01	0.230	0.818333
Walks	-1.064e+01	2.707e+00	-3.929	0.000110 ***
as.factor(Division)W	-9.646e+01	3.664e+01	-2.632	0.008996 **
CRBI:Hits	5.079e-03	1.416e-03	3.587	0.000401 ***
Hits:Walks	1.150e-01	2.158e-02	5.328	2.19e-07 ***
CRBI:PutOuts	3.833e-04	1.909e-04	2.008	0.045745 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 292.1 on 254 degrees of freedom

(59 observations deleted due to missingness)

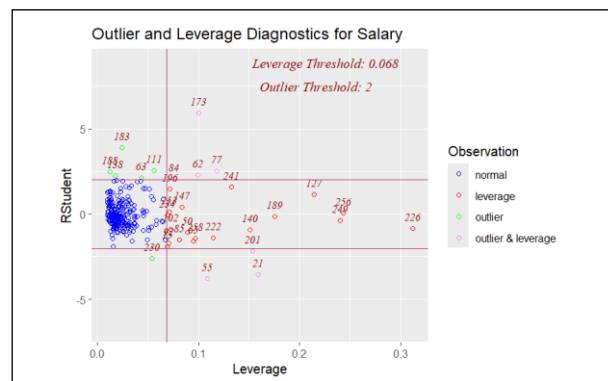
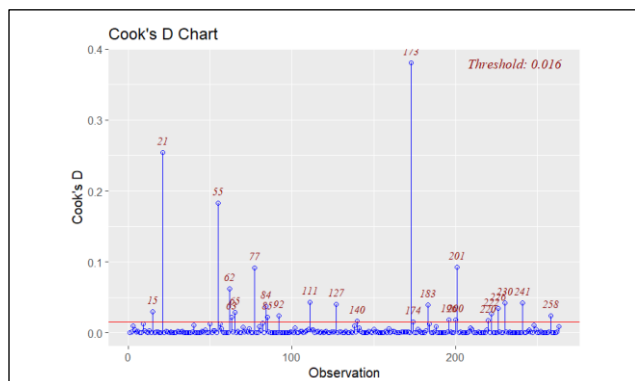
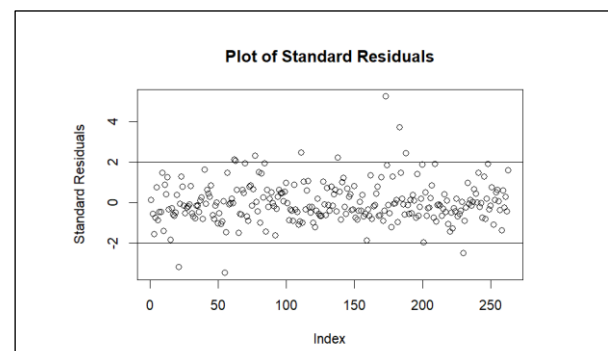
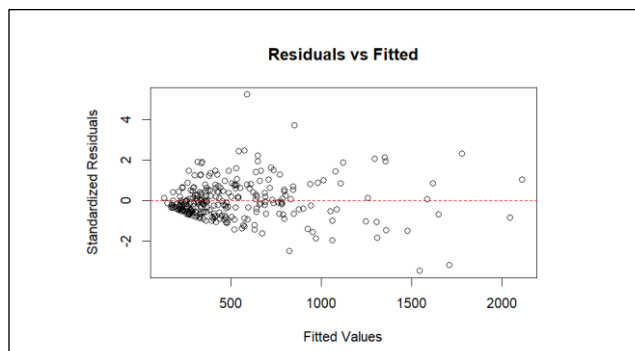
Multiple R-squared: 0.5935, Adjusted R-squared: 0.5807

F-statistic: 46.35 on 8 and 254 DF, p-value: < 2.2e-16

The finalized regression model provides a comprehensive equation for predicting a player's salary and the model explains 59.35% ($R^2 = 0.5935$) of the total variance in player salaries.

$$\text{Salary} = 591.4 - 0.006086 \cdot \text{CRBI} - 3.593 \cdot \text{Hits} + 0.02497 \cdot \text{PutOuts} - 10.64 \cdot \text{Walks} - 96.46 \cdot \text{DivisionW} + 0.005079 \cdot \text{CRBI} \cdot \text{Hits} + 0.1150 \cdot \text{Hits} \cdot \text{Walks} + 0.0003833 \cdot \text{CRBI} \cdot \text{PutOuts}$$

Checking the adequacy of the final model and identifying potential violations of the regression assumptions was done next. The plot of standardized residuals against fitted values revealed a distinct **wedge-shaped (heteroscedastic)** pattern, indicating that the variance of the error terms is not constant. This violation can lead to inefficient estimates and inaccurate standard errors. Furthermore, analysis of Cook's Distance and leverage values identified several influential points and potential outliers. Also, the presence of these points, combined with the heteroscedasticity, suggests that while the model is statistically significant, its predictive reliability could be improved through techniques such as a response variable transformation (e.g., using **log (Salary)**).



Refined multiple linear regression model (Using a log transformation)

Next, a refined multiple linear regression model was developed to predict the baseball player salaries, by utilizing a **log-transformed response variable** to address the heteroscedasticity issues identified in previous models. The model was constructed through a systematic forward selection process and with careful attention to multicollinearity and variable significance. This refined model did not reveal any significant interaction terms among the predictor variables.

$$\log(\text{Salary}) \sim \text{Hits} + \text{Years} + \text{PutOuts} + \text{Walks} + \text{CWalks} + \text{Division}$$

This model was developed through an iterative process that began with comprehensive pairwise analysis of all potential predictors against $\log(\text{Salary})$, followed by sequential forward selection, multicollinearity diagnosis using VIF, and incorporation of categorical variables through nested F-tests as was done previously.

The ANOVA results for this model demonstrate that Hits and Years are highly significant predictors of $\log(\text{Salary})$. PutOuts, Division and Walks are statistically significant. But CWalks adds negligible explanatory power ($F = 0.20831$, $p = 0.6485$), suggesting it could potentially be dropped for model parsimony. Hence, CWalks was removed from the refined model and the ANOVA table and summary table were obtained as shown below.

$$\log(\text{Salary}) \sim \text{Hits} + \text{Years} + \text{PutOuts} + \text{Walks} + \text{Division}$$

Analysis of Variance Table

Response: log(Salary)					
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Hits	1	41.871	41.871	107.5226	< 2.2e-16 ***
Years	1	57.992	57.992	148.9210	< 2.2e-16 ***
PutOuts	1	2.430	2.430	6.2401	0.013115 *
Walks	1	2.711	2.711	6.9616	0.008836 **
as.factor(Division)	1	2.069	2.069	5.3134	0.021960 *
Residuals	257	100.080	0.389		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Call:
lm(formula = log(Salary) ~ Hits + Years + PutOuts + Walks + as.factor(Division))
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.98610 -0.46300  0.06483  0.41328  3.12556
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.3167391  0.1263269  34.171  < 2e-16 ***
Hits         0.0063030  0.0010756   5.860  1.41e-08 ***
Years        0.0950303  0.0081503  11.660  < 2e-16 ***
PutOuts      0.0003072  0.0001459   2.106  0.0362 *
Walks        0.0057753  0.0022429   2.575  0.0106 *
as.factor(Division)W -0.1781438  0.0772834  -2.305  0.0220 *
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Residual standard error: 0.624 on 257 degrees of freedom
(59 observations deleted due to missingness)
Multiple R-squared:  0.5169,    Adjusted R-squared:  0.5075
F-statistic: 54.99 on 5 and 257 DF,  p-value: < 2.2e-16
```

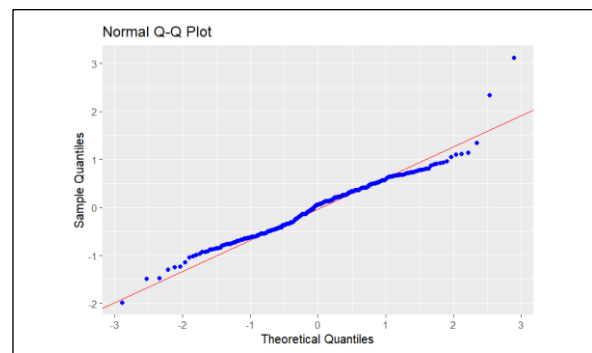
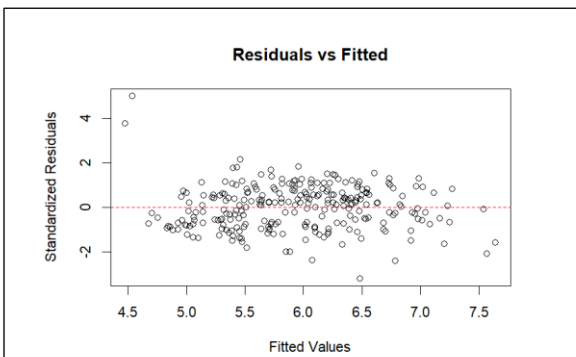
Thus, the finalized model provides a comprehensive equation for predicting a player's log transformed salary:

$$\log(\text{Salary}) = 4.3167 + 0.0063 \times \text{Hits} + 0.0950 \times \text{Years} + 0.0003 \times \text{PutOuts} + 0.0057 \times \text{Walks} - 0.1781 \times \text{DivisionW}$$

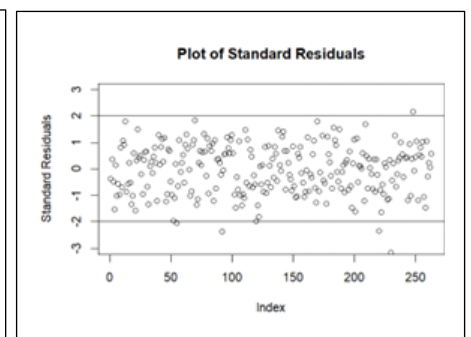
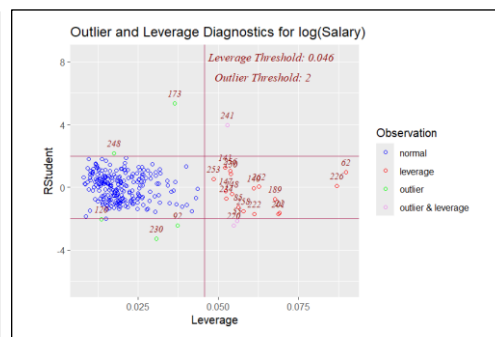
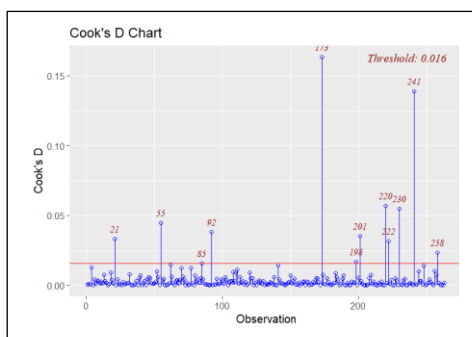
The finalized regression model provides a comprehensive equation for predicting the natural logarithm of baseball player salaries, representing a substantial improvement over previous

specifications through the log transformation of the response variable. This model demonstrates strong explanatory power, with a multiple R-squared value of 0.5169, indicating that approximately 51.7% of the variance in log-transformed salaries is accounted for by the included predictors. The overall model significance is confirmed by an F-statistic of 54.99 ($p < 2.2e-16$).

The diagnostic analysis confirms that the log transformation successfully resolved the heteroscedasticity issues present in the previous models. The residual plots now display a random scattered pattern without wedge shapes of irregular patterns, and the normal Q-Q plot indicates approximately normally distributed errors. These findings demonstrate that the model now satisfies the core assumptions of linear regression, providing reliable foundations for statistical inference.



The assessment of influential observations revealed no critical concerns. Cook's distance values identified only a few moderately influential points. Similarly, leverage analysis showed a small number of observations with elevated leverage, but none exerted disproportionate influence on the model parameters. Only a few standardized residuals indicated outliers beyond the ± 2 limits. This reinforces a generally good model fit across observations.



Multicollinearity diagnostics confirmed the model's stability, with all VIFs well below the conservative threshold of 10 (They lie around 1, indicating near orthogonality). This indicates that the predictor variables provide distinct information without problematic correlations. Collectively, these diagnostic results validate the model's adequacy for both explanatory and predictive purposes, with no substantial violations of the regression assumptions remaining.

```
> vif(parsimonyModel)
```

Hits	Years	PutOuts	Walks
1.584935	1.026987	1.121975	1.596353

In conclusion, the final regression model represents a robust statistical foundation for predicting baseball player salaries, successfully addressing earlier diagnostic concerns through the application of a log transformation and careful variable selection. The model demonstrates strong explanatory power while adhering to key regression assumptions, validated through comprehensive residual analysis, influence diagnostics, and multicollinearity assessment. With its balance of parsimony and predictive accuracy, this model provides reliable insights into salary determinants and offers a validated tool for both explanatory analysis and practical forecasting in player compensation evaluation.

Model Interpretation

$$\log(\text{Salary}) = 4.3167 + 0.0063 \times \text{Hits} + 0.0950 \times \text{Years} + 0.0003 \times \text{PutOuts} + 0.0057 \times \text{Walks} - 0.1781 \times \text{DivisionW}$$

1. Intercept: The predicted log (Salary) is 4.317 when all predictor variables are zero. This means a player with 0 Hits, 0 Years of experience, 0 Putouts, 0 Walks, and in the baseline division (assumed to be Division E) has a predicted salary of $\exp(4.317)$.
2. Hits: For each Hit, a player's log (Salary) value is predicted to increase by 0.0063 units, holding all other factors constant.
3. Years: For each additional Year of experience, a player's log (Salary) value is predicted to increase by 0.0950 units, holding all other factors constant. This is the strongest predictor in the model.
4. Putout: For each additional Putout, a player's log (Salary) value is predicted to increase by 0.0003 units, holding all other factors constant.
5. Walk: For each additional Walk, a player's log (Salary) value is predicted to increase by 0.0057 units, holding all other factors constant.
6. Division-W: Playing in Division "W" is associated with a predicted log (Salary) value that is 0.1781 units lower than playing in Division "E" (the reference group), holding all other player statistics constant.

Related R-Codes and R files can be downloaded from:

https://drive.google.com/drive/folders/1a8v1c0HSwuuAVHpsXJQk5D378a4X1V8i?usp=drive_link

Categorized response variable and Logistic regression

The response variable **Salary** was categorized into two distinct categories using the median salary as the base to categorize. The final categories were **high** and **low** based on whether the given salary is above or below the median salary respectively.

The dataset was then split into two with **training set** consisting of 80% of the data and **test set** consisting of 20% of the data. A stratified splitting procedure was implemented to ensure the ratio of "high" and "low" values were preserved in both train and test set

Model Fitting

A full model with step() function was used to build the initial model, then collinearity was observed and variables and interactions were dropped/added to arrive at the final model. The final model is as follows $Y \sim \text{AtBat} + \text{Hits} + \text{PutOuts} + \text{Walks} + \text{Runs} + \text{AtBat} * \text{CRuns}$.

Model Adequacy

Upon building the model the *test set* was used to draw a confusion matrix, the results are as follows

Confusion Matrix and Statistics

```

Reference
Prediction 0 1
0 24 7
1 0 21

Accuracy : 0.8654
95% CI : (0.7421, 0.9441)
No Information Rate : 0.5385
P-Value [Acc > NIR] : 5.751e-07

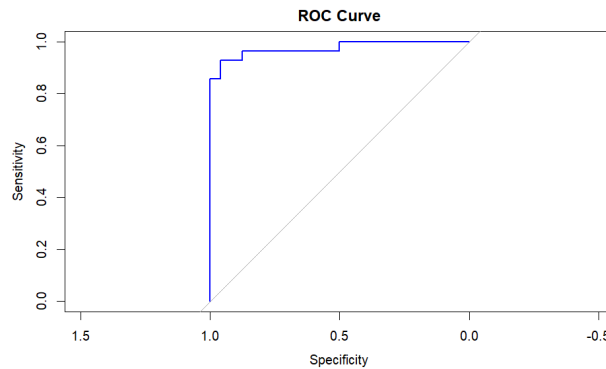
Kappa : 0.7347

McNemar's Test P-Value : 0.02334

Sensitivity : 0.7500
Specificity : 1.0000
Pos Pred Value : 1.0000
Neg Pred Value : 0.7742
Prevalence : 0.5385
Detection Rate : 0.4038
Detection Prevalence : 0.4038
Balanced Accuracy : 0.8750

'Positive' Class : 1

```



It is to be noted that values presented are subject to change based on the data in the *stratified training set*, and upon recreation of the experiment the output values would change, after multiple tests with parameters for explanatory variables, the previously mentioned model was reached.

The model at this specific instance had an accuracy of 0.8654 with 95% confidence interval being (0.7421, 0.9441). changing of the prediction class based on probability values was attempted, however $p = 0.5$ was robust, giving consistent results upon recreation of the experiment. The area under ROC curve was 0.9747 at this specific instance and it consistently reached above 0.90, ensuring the model fitted was a good fit, overfitting was reduced through multiple runs of the experiment.

Final Model

$$Y = -1.263 - 2.033e-02 * \text{AtBat} + 6.805e-02 * \text{Hits} - 3.860e-02 * \text{Runs} + 3.340e-02 * \text{Walks} + 1.326e-03 * \text{PutOuts} - 2.491e-03 * \text{CRuns} + 2.496e-05 * \text{AtBat} * \text{CRuns}$$

Where $Y = \ln(p/1-p)$

Discussion and Conclusions

This study was done to understand the factors that influence baseball player salaries using regression modeling. The analysis began using multiple linear regression model. After that improved by applying a log transformation to the response variable as a solution for the violation of the **constant variance assumption** (heteroscedasticity).

The initial model included **CRBI, Hits, PutOuts, Walks, and Division**, and some interaction terms also found significant. The estimated model equation was as bellow,

$$\text{Salary} = 591.4 - 0.006086 \cdot \text{CRBI} - 3.593 \cdot \text{Hits} + 0.02497 \cdot \text{PutOuts} - 10.64 \cdot \text{Walks} - 96.46 \cdot \text{DivisionW} + 0.005079 \cdot \text{CRBI} \cdot \text{Hits} + 0.1150 \cdot \text{Hits} \cdot \text{Walks} + 0.0003833 \cdot \text{CRBI} \cdot \text{PutOuts}$$

This model explained **59.35% of the variance in salaries ($R^2 = 0.5935$)**. However, diagnostic checks showed problems. First, multicollinearity was detected, but this was resolved after checking **VIF values**. The serious issue was a wedge-shaped pattern in the standard residuals vs fitted values plot, indicating **heteroscedastic** (non-constant variance). This made the estimates less reliable, and therefore a log transformation of Salary was applied.

The log transformed model addressed these issue and produced the following equation:

$$\log(\text{Salary}) = 4.3167 + 0.0063 \cdot \text{Hits} + 0.0950 \cdot \text{Years} + 0.0003 \cdot \text{PutOuts} + 0.0057 \cdot \text{Walks} - 0.1781 \cdot \text{DivisionW}$$

This final model explained **51.69% of the variance in $\log(\text{Salary})$** (Multiple $R^2 = 0.5169$, Adjusted $R^2 = 0.5075$). Important thing is the diagnostic plot confirmed that the constant variance assumption was now satisfied, meaning the model became more reliable for inference.

The Logistic regression model achieves an **AUC of above 0.9** and correctly classifies **86% of players** as high or low earners. The key predictors (Hits, PutOuts, Walks) are consistent across both models, which increases confidence in their importance. However, while the regression provides precise salary estimates, the logistic regression offers a simpler binary decision useful for classification tasks. Dichotomizing salary does lead to some information loss, but the classification accuracy is sufficiently high to be practically useful.

$$\text{Ln}(p/(1-p)) = -1.263 - 2.033e-02 \cdot \text{AtBat} + 6.805e-02 \cdot \text{Hits} - 3.860e-02 \cdot \text{Runs} + 3.340e-02 \cdot \text{Walks} + 1.326e-03 \cdot \text{PutOuts} - 2.491e-03 \cdot \text{CRuns} + 2.496e-05 \cdot \text{AtBat} \cdot \text{CRuns}$$

Summary of Findings

- Player salaries are influenced by **performance measures** (Hits, PutOuts, Walks), **experience (Years)**, and **Division**.

- The final model showed that **years of experience** was the strongest factor. It is reasonable, since experienced players are generally valued more than beginners
- The initial model had interaction effects. But these were not significant after the log transformation.
- The log transformation corrected the violation of constant variance and produced a stable model.
- The final model explains about **51.7% of salary variation**, showing good but not complete predictive power.
- The logistic Regression model has a high accuracy of (**74% - 94%**), with a very good and robust fit to the data

Conclusions

The final regression model provides a clear explanation of how salaries are determined. Both performance based measures and contextual factors like division significantly affect salary. **Experience** plays a key role. It is the strongest single predictor of salaries. The log transformed model offers a more accurate and reliable statistical framework compared to the untransformed model. The logistic regression model has a high accuracy in predicting the dichotomous salary category, but loses information compared to the linear regression model

Challenges and Limitations

- There were a few outliers and influential points in the dataset, and these might have had some effect on the results.
- The model only explains about half of the variations in salaries. This suggests that while performance and experience matter, there are also other factors not included in the dataset that could influence salaries.
- The analysis focused on linear regression, more complex methods might capture nonlinear or hidden relationships.

Even with these limitations, this analysis still gave useful insights into what affects baseball player salaries and showed how statistical modeling can help find patterns in real data.

Contributions

A.S Liyanage - s16948: Dataset Selection, Introduction, Literature Review & Editing

M.N.A Ahamed - s16974: Dataset Selection, Building Categorical response variable model, Advance Analysis

K.K.D.T.D Perera - s16954: Dataset selection, MLR Model Building and Interpretation, Refitting the model with the log transformation and Model Adequacy Checking, Advance Analysis

P.P. Jayasundara - s16806: Dataset selection, MLR Model Building, Discussion and Conclusion, Cover page Design

I.J. Kannangara - s16854: Dataset Selection, Descriptive analysis